

Worked Solution (Formula Solution)

Here is the cash flow timeline:

Year	0	1	2	3	4	5	6	...
Cash Flows (millions)	\$11.0	-\$8.9	-\$8.9	-\$8.9	\$5.4	\$3.78	\$2.65	

To compute the IRR, find the discount rate that sets the NPV equal to zero:

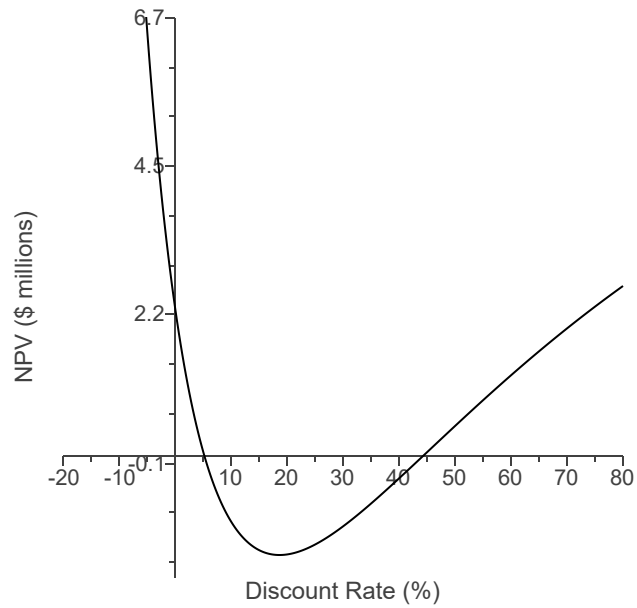
$$NPV = \$0 = \$11,000,000 - \frac{\$8,900,000}{r} \times \left[1 - \frac{1}{(1+r)^3} \right] + \frac{1}{(1+r)^3} \times \left[1 - \frac{\$5,400,000}{r+0.30} \right]$$

Therefore,

$$r = IRR = 5.30\% \text{ or } 44.12\%$$

One way to determine how many solutions this equation has is by plotting the NPV as a function of r .

Here's the graph:



Notice that the plot line crosses zero twice—once at 5.30% and again at 44.12%. There are two IRRs in this problem.

In this case, there are two IRRs (the intercepts on the x-axis) and the IRR does not provide an answer, so we cannot use the IRR rule. Therefore, the IRR rule does not work in this case.

